Reg. No. | | | | | | | | | | | [23MSCH1H04]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2024

ANALYTICAL CHEMISTRY

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. How is the confidence interval and the confidence level for samples expressed?

b. Describe the classifications of chromatographic techniques.

c. Discuss the application of solvent extraction.

d. What are the applications of thermogravimetry?

e. What is the Randle Servick equation? What are the assumptions of the equation?

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Discuss the solvent and column selectivity in RPHPLC?

b. Describe the theory of indicators with examples in acid-base and redox titrations.

c. Discuss the techniques of PCR and real time PCR.

SECTION – C

3. Answer the following question. **12**

a. i. Explain the different stages involved in chemical analysis.

ii. Define the following; 1) Retention time 2) Retention volume 3) Baseline width.

iii. How does one identify the reversibility and irreversibility of an electron transfer process by cyclic voltammetry ?